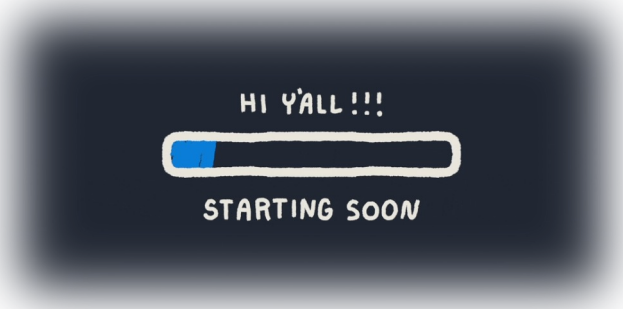
Hypothesis Testing

Trainer: Dr. Darshan Ingle





What are these statements? How are they proved or disproved?

- **GENERAL EXAMPLES**

- Earth is flat.
- Continents don’t move.
- 10000 hours of properly guided practice is the magic of greatness.
- Stress causes Ulcers.
- Men are better drivers than women.
- Earth is the center of the Universe.

What are these statements? How are they proved or disproved?

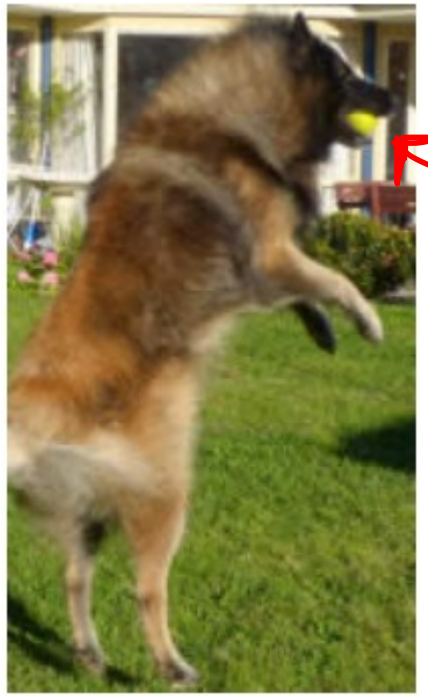
- **BUSINESS EXAMPLES**

- No difference in performance of the sales team across geographies and product lines.
- Change in gas price will have no impact on loses in automotive finance.
- Change in the CEO will have no impact on the stock price.
- Real estate yields are the same in all metros.
- Compensation changes will not impact attrition.

Hypothesis Testing

- A statement that might be true, which might then be tested.

  Example: Sam has a hypothesis that "large dogs are better at catching tennis balls than small dogs". We can test that hypothesis by having hundreds of different sized dogs try to catch tennis balls.



Importance of p-value

Perform this experiment :

1> flip a coin -> assume we get "head"

// P(H) = 0.5

2> flip the coin again -> assume we get "head" again

// P(H | H) = 0.5 x 0.5 = 0.25

Now before proceeding : "**Is this a fair coin ?**"

[What's your answer Yes / No ?]

[What-ever your ans ? How to you support your answer ?]

Here Null hypothesis = H_0 = Is a fair coin

Lets keep on tossing our coin to figure out whether we should

1> Reject H_0

2> or Not Reject H_0

Importance of p-value

3> flip the coin -> assume we get "head" again

P(H | 2H) = 0.5 x 0.5 x 0.5 = 0.125`

4> flip the coin -> assume we get "head" again

P(H | 3H) = 0.5 x 0.5 x 0.5 x 0.5 = 0.0625

That's 6% chance of we getting 4 heads in a row.

5> {last time} flip the coin -> assume we get "head" again

P(H | 4H) = 0.5 x 0.5 x 0.5 x 0.5 x 0.5 = 0.0312

That's 3% chance of we getting 5 heads in a row.

2 concepts :

Confidence level = 97%

and p-value or significance-level = 3%

We can say that chances of we getting a 5H in a row is just 3% or not getting 5H in a row is 97%.

Importance of p-value

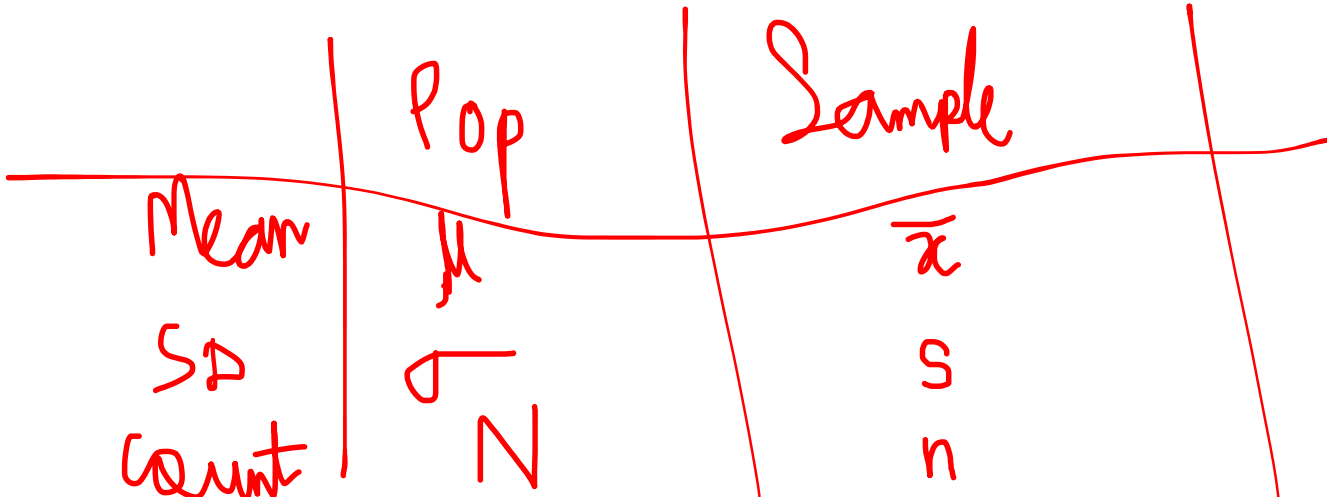
We can say that chances of we getting a 5H in a row is just 3% or not getting 5H in a row is 97%.

Hence we can Reject our Null Hypothesis H0 i.e. the coin is fair.
[**Note** : When we reject H0 we accept H1 i.e. the Alternative Hypothesis i.e. the coin is not fair !!]

- **Remember :** The p-value is a calculation that we make during hypothesis testing to determine if we reject the null hypothesis or fail to reject it.

Hypothesis Testing Examples

The average IQ for the adult population is 100 with a standard deviation of 15. A researcher believes this value has changed. The researcher decides to test the IQ of 75 random adults. The average IQ of the sample is 105. Is there enough evidence to suggest the average IQ has changed?



Hypothesis Testing

1. State null (H_0) and alternative (H_1) hypothesis
2. Choose level of significance (α)
3. Find critical values
4. Find test statistic
5. Draw your conclusion

Z / T

Formula

Step 1 : State Null & Alternative Hypothesis

$H_0: \mu = 100$

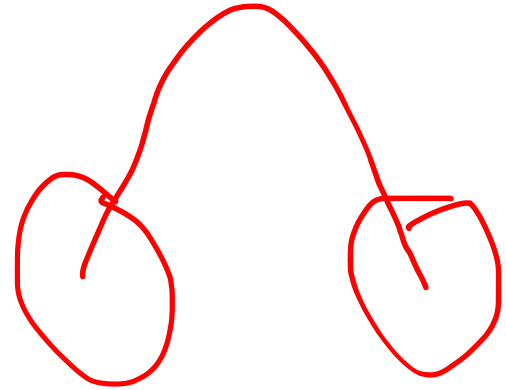
average IQ for the adult population is 100

$H_1: \mu \neq 100$

A researcher believes this value has changed.

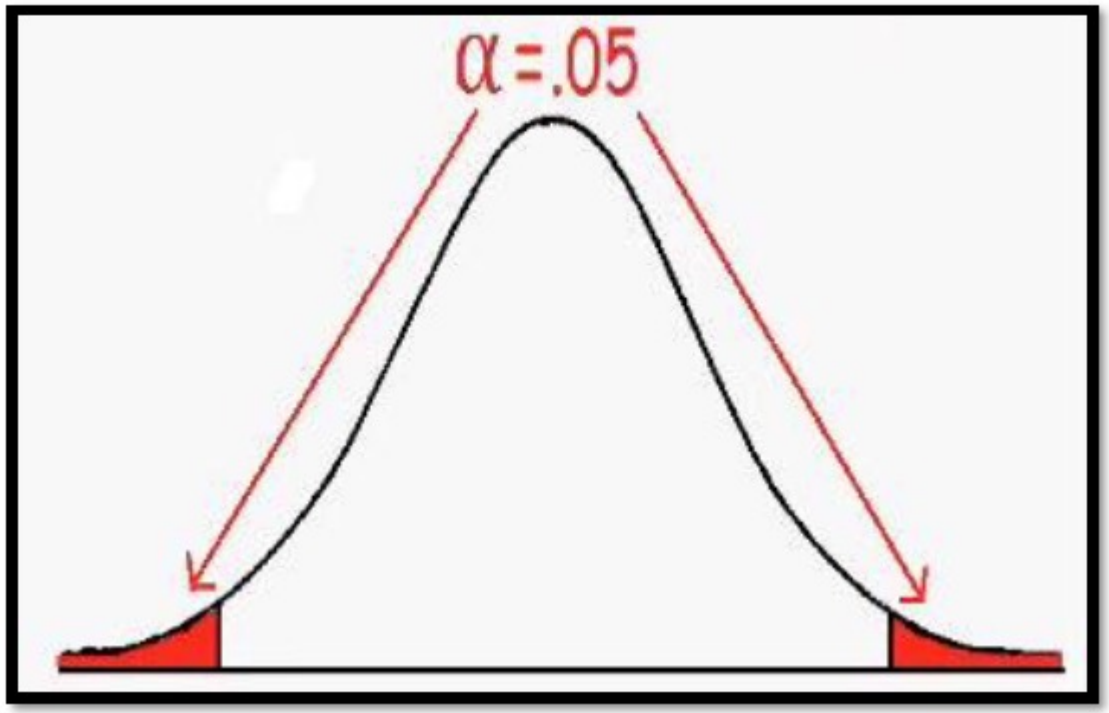
Simple Rules :

1. Null Hypothesis should always have = symbol
2. Alternative Hypothesis could have one of these > , < , >= , <= , !=
3. Alternative Hypothesis is **"what is being claimed"**
4. **When H_1 has != then we do a two-tailed test**
5. **When H_1 has either > , < , >=, <= then we do a one-tailed test**



Note : Hypothesis Testing is done to prove whether H_0 can be Rejected or Not be Rejected

Step 2 : Choose level of Significance



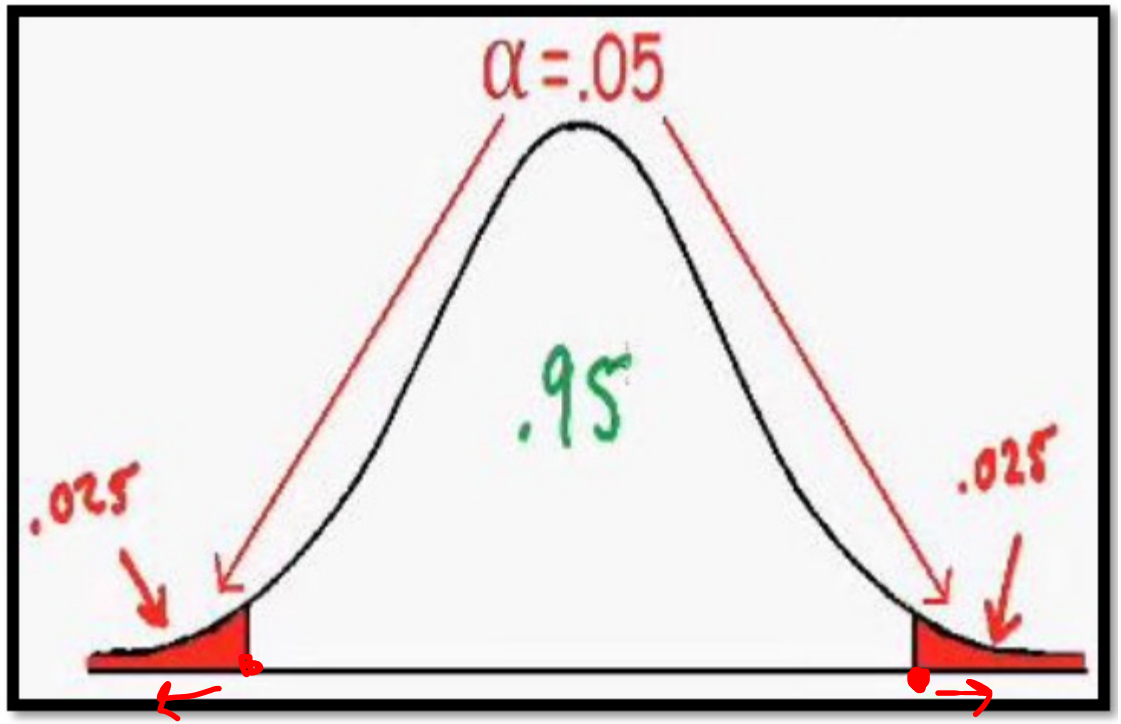
This is the area under the curve (**marked red**).

Usual values are 5% or 1%

5 % => .05

1 % => .01

Note : If our **test statistic** comes in **(red) tail part** then we would **Reject H_0** **else** we **will not Reject H_0**



Step 3 : find Critical Values

Here we need find z-value or t-value.

How to decide between z-value or t-value ?

1> If the **std deviation of the population** is not given. σ – Unknown

and **2>** If the **sample size < 30** n < 30

Then find a t-value , by using a t-table.

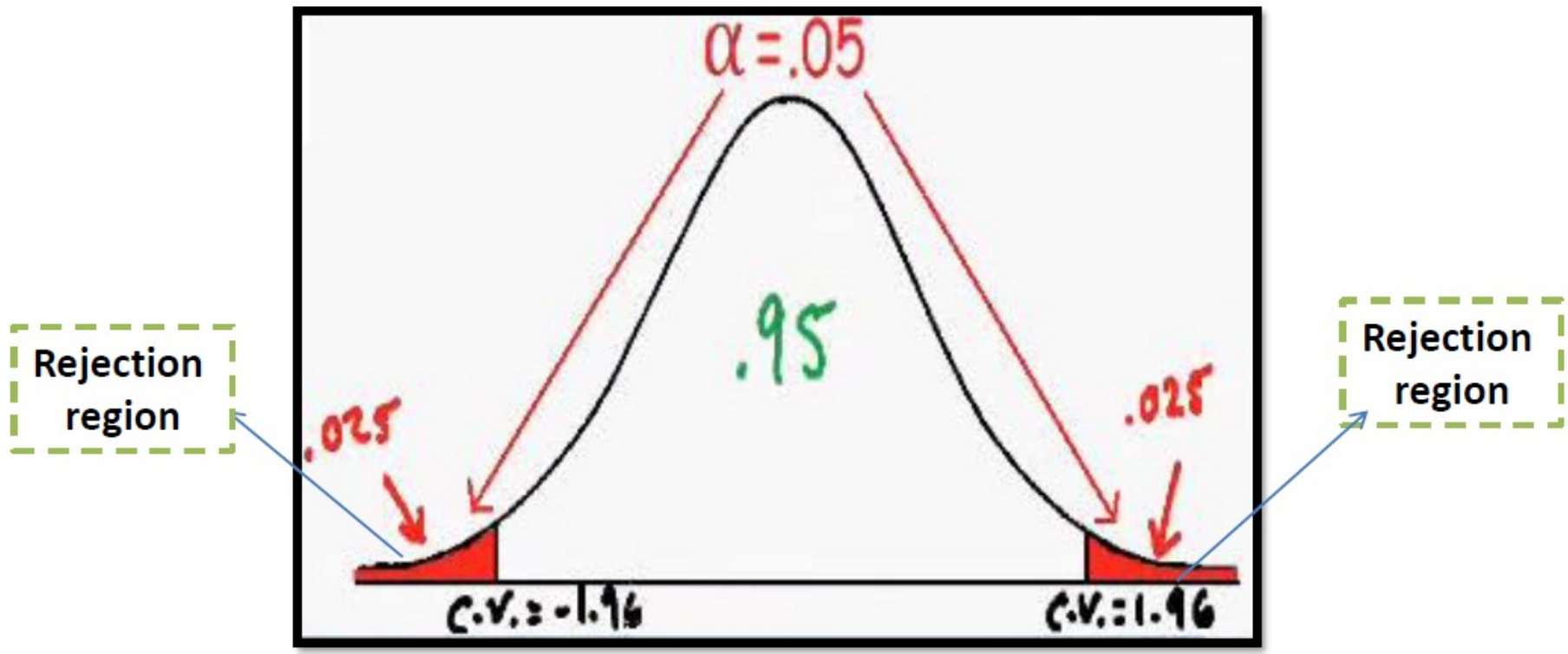
Z-table

Confidence Level	Area between 0 and z-score	Area in one tail (alpha/2)	z-score
50%	0.2500	0.2500	0.674
80%	0.4000	0.1000	1.282
90%	0.4500	0.0500	1.645
95%	0.4750	0.0250	1.960
98%	0.4900	0.0100	2.326
99%	0.4950	0.0050	2.576

$\sigma = 15$ standard deviation of 15.

sample size > 30

researcher decides to test the IQ of 75 random adults.



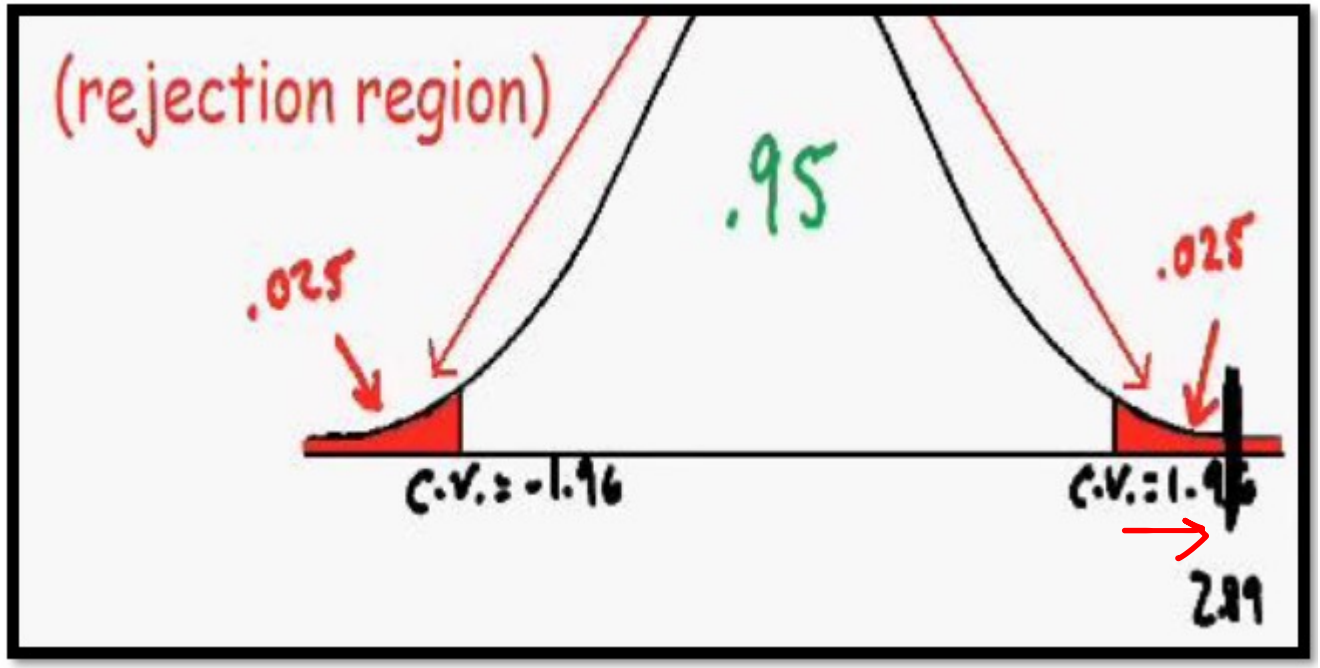
Step 4 : find test statistic. *i.e. the Z-value in this case*

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

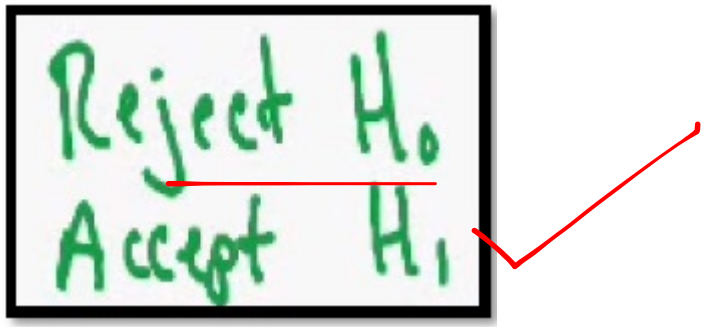
$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{105 - 100}{15/\sqrt{75}} = 2.89$$

test statistic

So our test statistic value is **2.89**



Step 5 : Conclusion



Hence, we say with 95 % confidence that there is enough evidence to support the claim : "Avg IQ has changed"